

Reduction of Oxidative Stress May Play a Role in the Anti-Inflammatory Effect of the Novel Herbal Formulation in a Rat Model of Hydrochloric Acid-Induced Cystitis

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Aims: We investigated the effect of the multi-herbal medicine, WSY-1075 in an animal model of hydrochloric acid (HCl)-induced cystitis. **Methods:** Rats were randomly assigned to three groups: sham-operated (control), HCl-induced only (HC), and HC treated with WSY-1075 (HC + WT). Oral administration of either distilled water (control, HC) or WSY-1075 (400 mg/kg) was continued for 4 weeks. In HC and HC + WT groups, cystitis was induced with 0.4 M HCl beginning on the 22nd day. Rats in each group underwent cystometry, and bladders were examined for evidence of inflammation and oxidative stress. **Results:** Treatment with WSY-1075 decreased the frequency of urination and reduced inflammation of the bladder tissue in a rat model of HCl-induced cystitis. Compared with the control group, the HC group showed severe chronic inflammatory and fibrosis signs, and the inflammatory grades significantly decreased following WSY-1075 treatment in the HC-WT group. The HC + WT group showed a markedly decreased expression of pro-inflammatory cytokines compared to the HC group. The level of malondialdehyde was significantly greater in the HC group compared to the control group, and it was significantly reduced in the treated (HC + WT) group. The levels of superoxide dismutase increased in the HC + WT group, which confirmed the anti-oxidant effect of WSY-1075. **Conclusions:** We suggest that reduction of oxidative stress may play a role in this anti-inflammatory effect. *Neurourol. Urodynam.* 34:86–91, 2015. © 2013 Wiley Periodicals, Inc.

Key words: anti-oxidants; cystitis; inflammation; interstitial

INTRODUCTION

Painful bladder syndrome/interstitial cystitis (PBS/IC) is a sterile (other than bacterial infection) bladder condition characterized by suprapubic and/or bladder pain related to bladder filling, accompanied by urinary frequency and urgency.¹ It is a bladder disease complex that the Society for Urodynamics and Female Urology defines as “an unpleasant sensation perceived to be related to the urinary bladder, associated with lower urinary tract symptoms of more than 6 weeks duration, in the absence of infection or other identifiable causes.”² Although the underlying cause of PBS/IC is still uncertain, several mechanisms have been proposed, including autoimmune responses, unidentified bacterial or viral infection, defects in the bladder lining, toxic substances in the urine, and neurogenic inflammation. It has also been reported that mast cell infiltration in the bladder wall is a prominent characteristic of PBS/IC, suggesting an association with inflammatory responses. Thus, an important aspect of PBS/IC treatment may be the control of inflammatory reactions for treating PBS/IC.³

Because there are few effective treatments available for PBS/IC,⁴ patients are often willing to seek alternative treatments. Patient interests in complementary and alternative medicine (CAM) therapies have been growing and are reflected in the increased use of CAM for treatment of PBS/IC. Nutraceuticals or chemicals that induce physiologic changes have become popular for managing PBS/IC patients^{5,6}; however, the efficacy and mechanism of these agents for treatment remain controversial.

In this study, we used modified Gongjin-dan, which contains cervi parvum Cornu (deer antler velvet), Korean red ginseng, Corni Fructus, *Angelica gigantis* Radix, *Lycii* Fructus, *Dioscoreae*

Rhizoma, *Citri* Pericarpium, *Gastrodiae* Rhizoma, *Agastachis* Herba, *Cassiae* cortex, *Scutellariae* Radix, and *Schisandrae* Fructus, and is a multi-herbal formula available in Korea for its antifatigue and antiaging effects. It has also been used as an alternative medicine for the treatment of various neurodegenerative disorders.⁷

Korean red ginseng (the steamed root of *Panax ginseng* C.A. Meyer, family Araliaceae) is a frequently taken oral remedy, especially in Asian countries. The major components of raw ginseng are ginsenosides, which contain an aglycone with a dammarane skeleton.⁸ Ginseng has been further shown to have anti-allergic effects^{9,10} and anti-oxidant effect.¹¹ *Corni fructus* has traditionally been used for mitigating tinnitus, improving impotence, and decreasing excessive urination.¹² It also has antioxidative effects.¹³ Sung et al.¹⁴ reported that aqueous extracts of *Corni fructus* have anti-inflammatory and analgesic activities in murine RAW 264.7 macrophage cells. Anti-inflammatory effects of an ethanol extract of *Angelica gigas* were investigated in a inflammation model, and the anti-

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inflammatory responses were demonstrated by blocking the tumor necrosis factor- α -nitric oxide pathway.¹⁵

During the tissue damage and inflammation, reactive oxygen species (ROS) such as hydrogen peroxide are generated endogenously by infiltration of macrophages and neutrophils.¹⁶ The covalent binding of free radicals to the receptors of membranes changes the unsaturated fatty acid/protein ratio and commences lipid peroxidation. Lipid peroxides are unstable and easily break down to form various products including aldehyde, such as malondialdehyde (MDA). Thus, the measurement of MDA amount reflects the degree of lipid peroxidation in tissues.^{17,18} Total superoxide dismutase (SOD) is the enzyme that catalyzes the dismutation of superoxide into oxygen and hydrogen peroxide and plays a critical role in the defense of cells against the toxic effects of oxygen radicals.

In a prior study, the importance of controlling inflammatory response to improve PBS/IC symptoms was emphasized.⁴ Therefore, we sought to investigate the effect of modified Gongjin-dan, named WSY-1075, with the measurement of tissue oxidative stress in an animal model of hydrochloric acid (HCl)-induced cystitis.

METHODS

Preparation of the Herbal Formula (WSY-1075)

The herbal formula used in our experiments was WSY-1075, a modified Gongjin-dan. The major ingredients in the herbal formula are obtained from six plants; 25% Corni Fructus, 25% *Angelica gigantis* Radix, 25% *Lycii* Fructus, 10% *Cervi parvum cornu*, 10% *Ginseng* Radix Rubra, and 5% *Cassiae cortex*. WSY-1075 was made with ingredients from Gongjin-dan which are known to have especially strong immune-modulating properties. A mixture of the dried seeds of six plants was extracted with boiling tap water (0.25 g/ml) for 3 h. The extracts were filtered and concentrated in vacuo and then lyophilized. A venture company, Korea Bio Medical Science Institute (KBMSI) Co. Ltd., which is developing oriental herbal medicines, developed this product as a health supplement.

Animal Groups and Treatment Protocol

Twenty-four 12-week-old female Sprague–Dawley rats were treated under a protocol approved by the Institutional Animal Care and Use Committee (CUMC-2012-0124-01) and handled according to NIH guidelines. Rats were divided equally into three groups ($n=8$ in each): sham-operated (control), HCl-induced only (HC), and HC treated with WSY-1075 (HC + WT). In each group, once-daily oral administration of either distilled water (control, HC) or WSY-1075 (400 mg/kg) was continued for 4 weeks (28 days). WSY-1075 was dissolved in 1 ml of distilled water and administered orally through an 8F red Rob-Nel catheter once a day. This is the first experiment using WSY-1075 for a rat model of HCl-induced cystitis. Therefore, the dosage of WSY-1075 and duration of treatment were selected by referring to the result of an experiment in which male SD rats of the same age were used as the nonbacterial prostatitis model.

Cystitis was induced with HCl using the method reported by Rivas et al.¹⁹ beginning on the 22nd day with a slight modification. The periurethral area of the rats was cleaned with iodophor solution, and the urethra was cannulated with a sterilized polyethylene catheter. In HC and HC + WT groups, the instillation process of 0.2 ml of 0.4 M HCl into the bladder using a sterile syringe after emptying the bladder by aspiration was repeated for seven days. HCl was instilled transurethrally

and kept for 4 min. The sham-operated group (control) had identical urethral catheterization and urine aspiration, except that 0.2 ml of sterile saline was instilled into the bladder. Prophylactic antimicrobial treatment was provided with cephazolin (7 mg/kg) given s.c. after the intravesical instillation.

Cystometrography

After 4 weeks, cystometrography (CMG) was performed in all rats. Rats were anesthetized using a subcutaneous injection of 1.2 mg/kg urethane. A suprapubic midline laparotomy was made to expose the bladder, and a 25 G needle connected to polyethylene tubing was inserted into the bladder through the bladder dome. The tubing was connected to a pressure transducer and a Harvard syringe pump via a three-way stopcock to record intravesical pressure and to infuse saline into the bladder. After emptying the bladder, CMG was performed using a saline infusion at a rate of 0.04 ml/min. Contraction interval and contraction pressure were recorded using a polygraph (Grass 7D, Grass Institute Co., Quincy, MA).

Bladder Histologic Evaluation

After the CMG procedure was completed, the bladders of all rats were excised and weighed. Bladder tissues were harvested and divided into two individual sections for the histologic evaluation and measurement of oxidative stress. One bladder section was fixed in 20% buffered formalin for 96 h and the other was frozen at -80°C . The bladder tissue was embedded in paraffin, cut into 4 μm transverse sections, and stained with hematoxylin–eosin. For each tissue section, five random fields were examined. Bladder inflammation was assessed by a pathologist in blinded fashion in the department of pathology in the catholic university of Korea, using a four-point scoring system: 0, morphologically unremarkable with no or minimal inflammation or epithelial changes; (1) mild inflammatory infiltrate within the lamina propria with scattered lymphocytes or monocytes, accompanied by mild chronic edema, hemorrhage, or urothelial changes; (2) moderate inflammatory infiltrate in the lamina propria and focal extension of the inflammation into the muscularis propria, accompanied by moderate chronic edema, hemorrhage, fibrin deposition, or urothelial changes; (3) severe inflammation in the lamina propria and muscularis propria in association with other significant findings, such as urothelial ulceration, severe chronic edema, hemorrhage, and fibrin deposition.

Cytokine Analyses

Blood was collected via blind cardiac puncture through the thoracic wall prior to sacrificing the rats. Blood samples were centrifuged for 10 min (3,000 rpm, 4°C) and the plasma was stored at -80°C until analysis. Cytokine concentration was measured using an immunoassay ELISA kit (R&D Systems, Minneapolis, MN) according to the manufacturer's protocol every 5 min for 30 min, using a spectrophotometer at 450 nm.

Measurement of Oxidative Stress

SOD activity (CuZnSOD and Mn SOD) in bladder tissue was measured using a SOD Assay Kit-WST (Dojindo), monitoring the decrease in the rate of superoxide-mediated reduction of nitroblue tetrazolium at 450 nm using a spectrophotometer. MDA assessment from peroxidation reaction products by free radicals was performed with the double heating method by

Draper and Hadley. The method is based on the principle of producing maximum absorbance at a 532-nm wavelength by an MDA reaction, which is the final product in fatty acid peroxidation with thiobarbituric acid (TBA).²⁰

Statistical Analysis

All data are presented as the mean $\pm$ SD. Statistical significance was evaluated using Kruskal–Wallis tests and Mann–Whitney *U* tests with the Bonferroni correction. Spearman's test was used for correlation analysis among CMG differences, inflammatory degrees, measurements of oxidative stress, and cytokine levels. In all cases, *P* values <0.05 were considered significant.

RESULTS

CMG Differences

Table I describes CMG parameters, including frequency, maximal detrusor pressure, contraction interval, bladder capacity, resting pressure, and threshold pressure. Figure 1 shows the representative CMG tracings. Contraction intervals and bladder capacity significantly decreased in the HC group compared with the control group ($P < 0.05$). The decrease in these variables induced by intravesical HCl was significantly suppressed by WSY-1075 treatment ($P < 0.05$). However, there was no statistically significant difference in the resting pressure or threshold pressure among the three groups.

Bladder Inflammatory Changes

In the control group, there was a normal appearance of the mucosa with no obvious leukocyte infiltration, resulting in assignment of inflammatory grade of 0–1 in all rats. Compared with the treatment group, the HC group showed more severe chronic inflammatory and fibrosis signs (Fig. 2), and the rats were assigned much higher inflammatory grades ($P < 0.001$; Table II). Diffuse monocytes or lymphocytes infiltrated both the lamina propria and muscularis propria, and small vascular proliferation was also seen in the lamina propria. The inflammatory grades were significantly decreased after WSY-1075 treatment ($P < 0.001$; Table II). Mild epithelial dysplasia was present in large areas, and dysplasia was present focally. However, edema and submucosal hemorrhage were markedly decreased.

Cytokine Analyses

To investigate the effects of WSY-1075 on the cystitis-induced production of the inflammatory cytokines interleukin (IL)-6, IL-

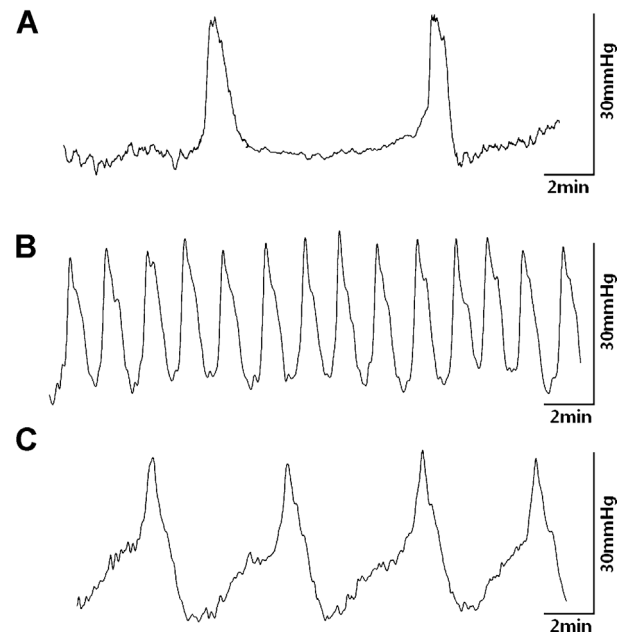


Fig. 1. Representative tracings showing the changes in voiding function during continuous infusion cystometry in the control group ($n = 8$) (A), the HC group ($n = 8$) (B), and the HC + WT group ($n = 8$) (C). HC, HCl-induced only; HC + WT, HC treated with WSY-1075.

8, and TNF- α , ELISA analyses were conducted. As shown in Figure 3A, a significant increase in pro-inflammatory cytokines was detected when the HC group was compared to the control group ($P < 0.05$). There was also a significant cytokine decrease in the HC + WT group compared to HC group ($P < 0.05$).

Comparison of Oxidative Stress in Bladder Tissues

There was a significant decrease in SOD levels in the HC group compared to the control group ($P = 0.035$); however, SOD levels in the treatment group were increased compared with HC group ($P = 0.04$) (Fig. 3B). MDA levels were significantly increased in bladder tissues of all rats in the HC group ($1.14 \pm 0.18 \mu\text{mol/mg protein}$) compared with the control group ($0.18 \pm 0.04 \mu\text{mol/mg protein}$) ($P = 0.002$). On the other hand, WSY-1075 treatment decreased the inflammation-mediated elevation in MDA levels ($0.73 \pm 0.05 \mu\text{mol/mg protein}$) to the control levels in HC + WT group ($P = 0.007$) (Fig. 3B).

DISCUSSION

The main findings of the present study were as follows: (1) treatment with the multi-herbal medicine, WSY-1075, decreased the frequency of urination by minimizing inflammation of the bladder tissue in a rat model of HCl-induced cystitis; (2) treatment with WSY-1075 decreased the expression of pro-inflammatory cytokines; and (3) levels of SOD increased and levels of MDA decreased following WSY-1075 treatment, which confirms the anti-oxidant effects of WSY-1075.

PBS/IC is a chronic painful disease of the bladder, and its prevalence is high. Traditional treatment options include oral medication, intravesical therapy, surgery and others.²¹ However, the treatment for patients with PBS/IC is a relatively challenging topic for urologists, because the exact etiology of this disease is still unclear. Ironically, the most frequent sources

TABLE I. Results of Cystometric Analysis of the Experimental Groups

	Control	HC	HC + WT
Frequency (contraction/hr)	6.7 $\pm$ 2.3	31.0 $\pm$ 2.1 ^a	16.5 $\pm$ 1.8 ^b
Pves Max at CMG (cmH ₂ O)	39.0 $\pm$ 4.1	36.5 $\pm$ 3.3	37.5 $\pm$ 6.5
Contraction interval (sec)	557.0 $\pm$ 13.1	105.1 $\pm$ 19.3 ^a	420.5 $\pm$ 22.8 ^b
Bladder capacity (ml)	1.2 $\pm$ 0.1	0.2 $\pm$ 0.1 ^a	0.8 $\pm$ 0.1 ^b
Resting pressure (cmH ₂ O)	2.5 $\pm$ 0.1	1.8 $\pm$ 0.5	2.1 $\pm$ 0.4
Threshold pressure (cmH ₂ O)	3.9 $\pm$ 0.3	4.1 $\pm$ 0.5	4.3 $\pm$ 0.8

HC, HCl-induced only; HC + WT, HC treated with WSY-1075.

^aSignificant statistical difference ($P < 0.05$) compared to the control.

^bSignificant statistical difference ($P < 0.05$) compared to HC group.

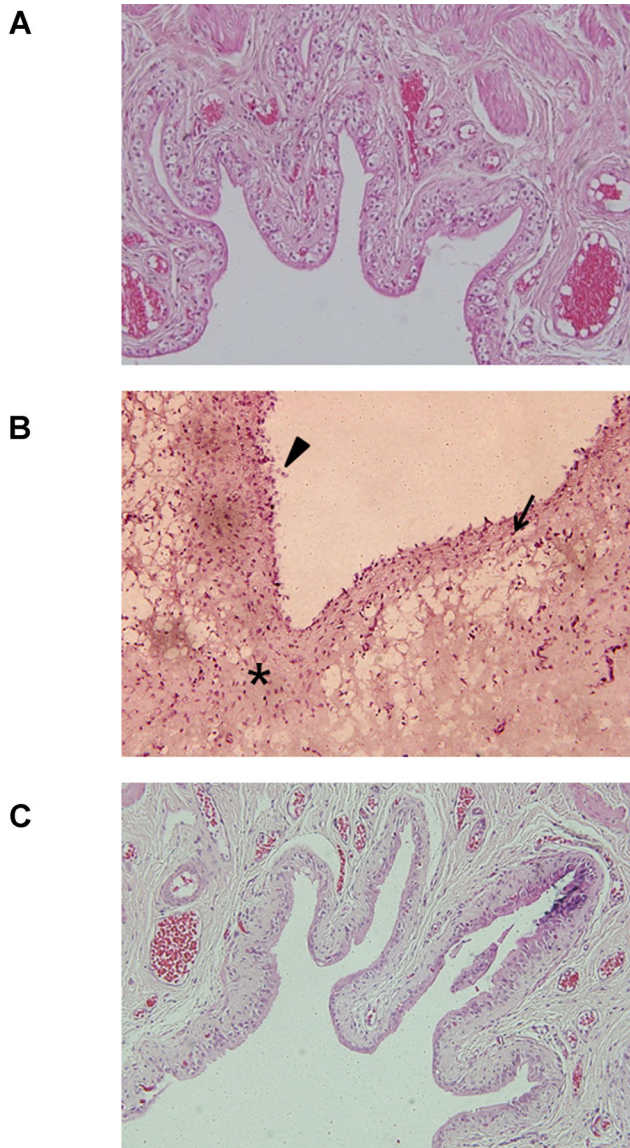


Fig. 2. Representative sections of hematoxylin–eosin stained in the bladders (200×) from the control group ($n=8$) (A), the HC group ($n=8$) (B), and the HC + WT group ($n=8$) (C). Diffuse infiltrated monocytes or lymphocytes (*), fibrosis (arrow) and epithelial denudation (arrow head) are shown in the HC group. HC, HCl-induced only; HC + WT, HC treated with WSY-1075.

of recommendations for the use of CAM in PBS/IC patients have been the patients' doctors. This may demonstrate not only physicians' acceptance of diet therapy as mainstream PBS/IC treatment, but also their response to increased patient interest, their own interest, or their willingness to try CAM because

TABLE II. Comparison of Bladder Inflammation

Groups	Inflammation(grade)				Body weight (g)	Bladder weight (mg)
	0	1	2	3		
Control	7	1	0	0	308.67 ± 13.32	145.83 ± 5.25
HC	0	1	2	5	313.67 ± 20.43	180.24 ± 10.08
HC + WT	0	4	3	1	311.75 ± 8.30	171.38 ± 22.24

HC, HCl-induced only; HC + WT, HC treated with WSY-1075.

traditional therapies have not been successful.²² A number of dietary supplements and herbs have been the subjects of research for PBS/IC treatment, including calcium glycerophosphate, L-arginine, Chinese herbs, and a quercetin-containing combination.^{3,23–26} However, only one supplement,²⁴ calcium glycerophosphate, was found to be helpful by a significant number of patients.

Herbal formulas have also been used as the treatment alternatives for PBS/IC. A pilot study utilized the herbs Cornus, gardenia, curculigo, rhubarb, and Rhemannia in a tea for 6 days a week for 3 months. Of the 25 patients, 61% had a significant decrease in pain at 4 weeks.²³ Although the previous study did not report functional data for the bladder, Lee et al.²⁷ demonstrated that the administration of the herbal medicine, Yukmijihwangtang might be an applicable therapeutic traditional medicine for the treatment of PBS/IC. Administration of Yukmijihwangtang mitigated histological changes of bladder and suppressed IL-6/TNF- α production in a rat model of chronic cystitis.

It was proposed that sensitization of bladder afferent pathways might be involved in the mechanisms inducing bladder pain and urinary frequency in PBS/IC.²⁸ Several reports demonstrated that pro-inflammatory agents induced functional changes in C-fiber afferents that can lead to these relatively unexcitable afferents becoming hyperactive or hyperexcitable.^{29–31} Similarly, cytokines and inflammatory mediators are known to be involved in sensitization of bladder afferent pathways under inflammatory conditions.²⁸ In the present study, we found that oral WSY-1075 treatment can decrease urinary frequency and histopathological changes in rats with HCl-induced cystitis. Furthermore, there was a positive correlation between inflammatory cytokine levels and the observed degree of inflammation. Thus, it is likely that the therapeutic effects of WSY-1075 are at least in part attributable to suppression of inflammatory responses and subsequent activation of bladder afferent pathways induced by cytokines or pro-inflammatory mediators.

Sunwoo et al.⁷ demonstrated that gongjin-dan inhibits inflammatory cytokines such as Cox-2, IL-1, and TNF- α in a cerebral ischemic model. IL-6 is a pro-inflammatory cytokine produced by various cell types, including endothelial cells, macrophages, and mast cells.³² It is important to the transition between acute and chronic inflammation and exhibits a pro-inflammatory profile in chronic inflammation.³³ The role of IL-6 in PBS/IC is controversial; however, Lv et al.³⁴ investigated the correlation between IL-6 levels and the degree of inflammation, as well as the voiding frequency, in protamine sulfate-induced chronic cystitis rats with or without treatment with hyaluronic acid. The elevation of a variety of inflammatory factors and the presence or activation of mast cells have been detected in cystitis animal models. Rats receiving cyclophosphamide treatment showed elevation of a majority of cytokine levels (including IL-4, IL-6, IL-18, TNF- α , and others).³⁵ Boucher et al.³⁶ concluded that mast cell presence and TNF- α expression can be used to evaluate the change and severity of bladder inflammation in nonbacterial cystitis model.

Tissue injury and inflammation, as well as many chronic and degenerative diseases, have strong ties to oxidative stress, which occurs when the rate of ROS generation during normal/aerobic metabolism exceeds the detoxification abilities of the cell.³⁷ The fact that there was a decrease in tissue MDA levels and a increase in tissue SOD levels after treatment in our study demonstrates the oxidative stress-relieving anti-oxidant effect of WSY-1075. Bayrak et al.³⁸ also studied the effects of intravesical dexpantenol use on bladder histology and lipid peroxidation in a chemical cystitis animal model. They used

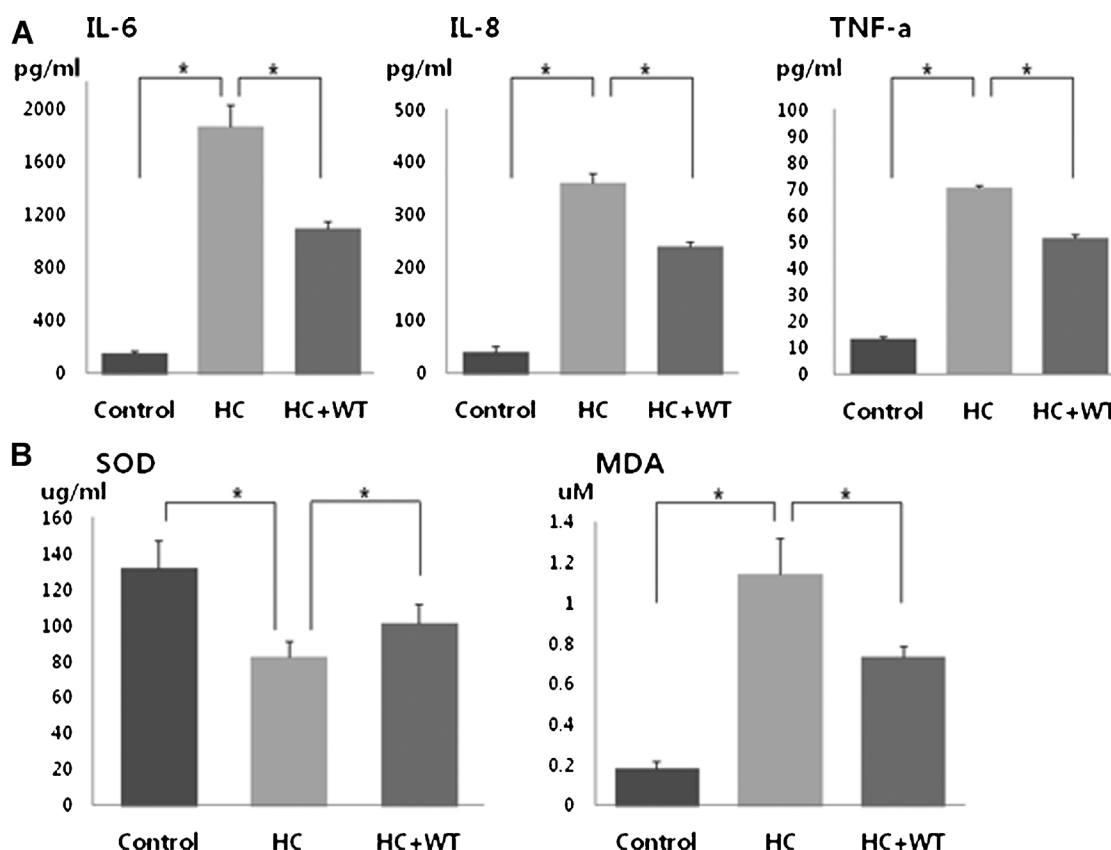


Fig. 3. Comparison of cytokines and expression levels of oxidative stress in bladder tissue from the control group ($n=8$), the HC group ($n=8$), and the HC + WT group ($n=8$). **A:** ELISA of IL-6, IL-8, and TNF- α (pg/ml). **B:** Total superoxide dismutase and malondialdehyde expression in bladder tissue. HC, HCl-induced only; HC + WT, HC treated with WSY-1075 * $P < 0.05$.

dexpanthenol in experimental chemical cystitis model because of its anti-inflammatory, wound healing, epithelization accelerating properties, and found that dexpanthenol decreased the MDA levels of serum and tissue, which are the products of lipid peroxidation, to a statistically significant extent. However, indeed, there is a limitation of applying intravesical instillation as a treatment modality, when compared with oral therapy.

The current study is the first to suggest that WSY-1075 has anti-inflammatory and anti-oxidant effects in a rat model of hydrochloric acid-induced cystitis. However, a potential limitation of our study is that we did not demonstrate the mechanism of each ingredient. Moreover, through this study, we have identified the anti-oxidant effect and anti-inflammation of the new herbal formulation in chemically induced cystitis, however we did not see a complete improvement of cystitis. Future work should investigate a detailed mechanism of WSY-1075 with its anti-oxidant effect in PBS/IC bladders. The use of WSY-1075 prior to development of cystitis rather than after development of cystitis, is not the usual manner in which patients would present for treatment. The bladder damage of the animal models used in the experiment was too severe that it was unlikely to be reversible. And since there was no prior experiment in regard of the effect we expected, we have used WSY-1075 prior to development of cystitis to minimize any irreversible damage.

In conclusion, the present study indicated that rats with HCl-induced cystitis had urinary frequency and pathological changes in the bladder, which are often seen in human PBS/

IC bladders. WSY-1075 decreased bladder inflammation severity and reduced levels of inflammatory cytokines in a rat model of PBS/IC. We suggest that reduction of oxidative stress may play a role in the anti-inflammatory effect.

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